XXX Olimpiada Informática Española

OIE 2026 — Final Día 1

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Organizan y financian

Patrocinan

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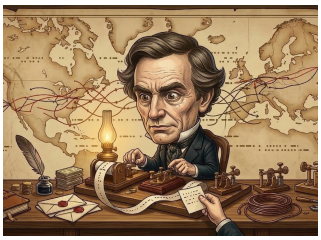
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Dots and Dashes

MORSE



Morse code is a system for representing characters using short and long signals, traditionally called *dots* (`.`) and *dashes* (`-`). It was developed in the 19th century for use in the telegraph and made it possible to transmit information over long distances using sequences of pulses.

In the standard version of Morse code, each decimal digit from 0 to 9 is represented by a combination of exactly five symbols, according to the following table:

Digit	Morse code		Digit	Morse code
0	`-----`		5	`.....`
1	`.----`		6	`-....`
2	`..---`		7	`--...`
3	`...--`		8	`---..`
4	`....-`		9	`----.`

To encode a sequence of digits, the representations corresponding to each digit are concatenated without adding any additional separators.

Given a sequence of digits, we want to know how many dots and how many dashes there are in total in its Morse code encoding.

Entrada y salida

The input begins with an integer T, which indicates the number of test cases.

Each of the following T cases consists of a line with a non-empty sequence s of decimal digits (`0`–`9`).

For each test case, print a line with two integers separated by a space: the total number of dots followed by the total number of dashes when encoding the sequence s in Morse code.

Restricciones

- $1 \leq T \leq 1000$
- The length of the sequence s ranges from 1 to 1000.

Subtareas

1. (15 points) The sequence s only uses the digit `0`.
2. (20 points) The length of the sequence s is 1.
3. (25 points) All digits in the sequence s are equal.
4. (40 points) No additional constraints.

Ejemplo

Entrada	Salida
3 5 10 314159	5 0 1 9 15 15

Channel with Interference

OPERATION

Professor Oak has sent a message through a channel with too much interference. This message consists of a sequence s of n *bits* (0 or 1), $s_1\ s_2 \ldots s_n$, which will take k milliseconds to reach its destination. Due to this interference, every millisecond the message undergoes a transformation such that s becomes s', where $s'_i = s_{i-1} \oplus s_{i+1}$[†]. Here indices are taken cyclically, i.e., $s_0 = s_n$ and $s_{n+1} = s_1$.

Can you predict what the final message will be that reaches the recipient after k milliseconds?

Entrada y salida

The first line of the input contains the number of test cases T.

Each test case consists of two lines. The first line contains two integers, n and k, the length of the sequence s and the number of milliseconds it takes to reach the recipient, respectively. The second line contains the sequence s of n bits.

For each test case, print the result of transforming s after k milliseconds of interference.

Restricciones

- $1 \leq T \leq 3 \cdot 10^4$
- $3 \leq n \leq 10^5$
- The sum of n over all test cases is at most 10^5.
- $1 \leq k \leq 10^{18}$
- $s_i \in \{0, 1\}$

Subtareas

1. (13 points) Both the sum of n and the sum of k over all test cases are at most 1000.
2. (28 points) $k \geq 10^{15}$, n is a power of 2 ($n = 2^r$ for some integer r).
3. (22 points) The sum of n over all test cases is at most 1000.
4. (37 points) No additional constraints.

Ejemplo

Entrada

```
3
7 1
1001111
3 1000
101
5 2
11011
```

Salida

```
1111000
101
10001
```

[†]The XOR operation $\oplus$ (exclusive OR) of two bits follows the table:

x_1	x_2	$x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0

Corrupted Backup

UNICOS



The central server of the OIE has suffered a slight data corruption in its backup system. In one of the memory sectors, there is a secret sequence a of $2 \cdot n + 1$ integers with values between 1 and n. By design, each value should be backed up by appearing exactly 2 times, but a *glitch* has caused a single value to appear exactly 3 times.

Your objective as a systems auditor is to find the exact 3 positions of this corrupted datum.

To avoid triggering the security protocols, you cannot download the entire sequence. Instead, you must use an API diagnostic function by making queries of the following kind:

1. Choose an integer k and a sequence of indices of size k, $s_1, s_2, \ldots, s_k$, with **distinct** values between 1 and $2 \cdot n + 1$.
2. You will receive as an answer the number of values in $\{a_{s_1}, a_{s_2}, \ldots, a_{s_k}\}$ that appear **exactly once**. We call these **unique values**. For example, if the values $a_{s_1}, \ldots, a_{s_k}$ were $\{2, 1, 2, 3, 2, 3, 6, 7\}$, the answer would be **3**, since 1, 6, and 7 are the only unique values. The 3 appears twice and the 2 appears three times, so they are not taken into account.

Interaction

Each execution begins with an integer T, the number of cases to process. In each case, communication with the judge follows this protocol:

- First, you must read an integer n, indicating that the hidden sequence has $2 \cdot n + 1$ integers.
- Next, you may perform queries. To query the sequence of indices $s = \{s_1, s_2, \ldots, s_k\}$, you must print `? k` s_1 s_2 ... s_k, where **the values in s must be pairwise distinct**.
- Immediately after each query, you will receive an integer: the judge's answer to your query.
- If you read `-1` after a query, it means you have exceeded the maximum number of queries or that your query is invalid. **Your program must terminate immediately** to receive the verdict "Wrong Answer" in CMS. Otherwise, your program may receive arbitrary verdicts.
- Once you have solved the test case, you must print the 3 special indices (the 3 positions of the corrupted datum) in any order using the format `!` i_1 i_2 i_3.
- After printing the result for the case, you will receive a `1` if your answer is correct and a `-1` if your answer is incorrect or invalid. If you receive a `1`, you may proceed to solve the next test case (if one exists). But if you receive a `-1`, **your program must terminate immediately** to receive the verdict "Wrong Answer" in CMS. Otherwise, your program may receive arbitrary verdicts.

Remember that you must flush the output every time you print data (using `cout << endl` or `cout << flush` in C++, `System.out.flush()` in Java, `sys.stdout.flush()` in Python).

For all test cases within the same execution (T), you have a global limit of at most $100 \cdot T$ queries. You may exceed 100 queries in a particular case as long as you compensate by using fewer queries in others.

The secret sequence a is fixed before each test case begins, that is, the judge is **not adaptive**.

Subtareas

1. (12 points) $T = 1$ and $n = 3$. You will get the points if you make no more than 100 queries.

2. (9 points) $T = 1$ and $n = 6$. You will get the points if you make no more than 100 queries.
3. (79 points) $T = 100$ and in all cases $n = 100$. You will get points according to f, which depends on the average number of queries q over all cases, defined as follows:

$$f(q) = \max(0, \min(79, 2 \cdot (63 - q))$$

To obtain full points, the problem must be solved in at most $23.5 \cdot T$ queries.

The test cases for this problem have been generated uniformly at random and for each subtask there is only a single batch of cases.

A (fictional) example of a test case with $T = 1, n = 2, a = [1, 2, 2, 1, 2]$ could be:

Judge	Contestant	Comment
1		value of T
2		value of n
	`? 3 1 2 3`	
1		Among the values $a_1 = 1$, $a_2 = 2$, and $a_3 = 2$, there is only one unique value: a_1.
	`? 1 4`	
1		
	`? 3 2 3 5`	
`0`		In $\{a_2, a_3, a_5\}$ there are no unique values.
	`! 2 3 5`	The positions in a with a value repeated 3 times are 2, 3, and 5
1		The case has been solved correctly.

Thieves in the Museum

LADRONES



In the Colegio de San Gregorio there is one of the branches of the National Sculpture Museum, a museum with n rooms connected to each other by doors. From any room it is possible to reach any other room, potentially having to pass through other rooms along the way, and, to prevent tourists from getting lost in the museum, it contains no cycles, that is, it has a tree structure.

It is known that there are several thieves in the museum and, using the minimum possible number of guards, we want to arrest all of them.

To arrest them, the following strategy is used to explore the museum:

- All guards start in room 1, where the exit is located.
- The guards move through different rooms in a coordinated way until all thieves have been arrested.

However, the thieves can also move from room to room as long as they do not encounter a guard. To prevent them from escaping, it is required that every room that has been checked and is connected to at least one room that has not been checked contains at least one guard.

What is the minimum number of guards needed to perform this task if the guards visit the rooms optimally?

Entrada y salida

The first line contains an integer T, the number of cases to process.

The first line of each case contains n, the number of rooms.

The next $n-1$ lines contain two integers u, v, indicating that there is a door between rooms u and v.

For each case, print an integer: the minimum number of guards needed to perform this task.

Restricciones

- $1 \leq T \leq 10^5$
- $1 \leq n \leq 10^5$
- $1 \leq u, v \leq n$ and the graph is a tree (a connected graph without cycles).
- The sum of n over all cases will be at most 10^5.

Subtareas

1. (16 points) All nodes in the tree have degree at most 2.
2. (11 points) The tree is a star (there exists a node such that all other nodes are at distance 1 from it).
3. (45 points) The tree has at most 8 leaves, $T = 1$ and $n \leq 100$.
4. (28 points) No additional constraints.

Ejemplo

Entrada

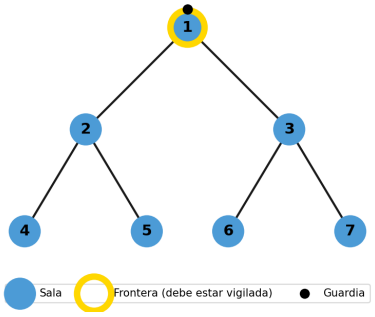
```
4
3
1 2
2 3
5
1 2
1 3
1 4
1 5
5
1 2
2 3
4 2
5 1
7
1 2
1 3
2 4
2 5
3 6
3 7
```

Salida

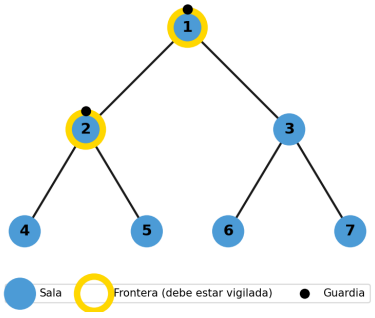
```
1
2
2
3
```

In the **fourth sample case**, the museum has the shape of a binary tree with 7 rooms. To catch the thieves using a minimum of **3 guards**, an optimal step-by-step strategy is the following:

- **Step 1:** All guards start in room 1 (exit). Since it is connected to unchecked rooms, it temporarily acts as the frontier.

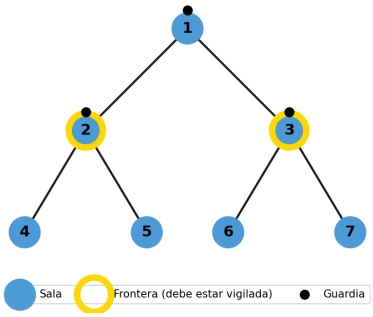


- **Step 2:** The guards move forward toward room 2...

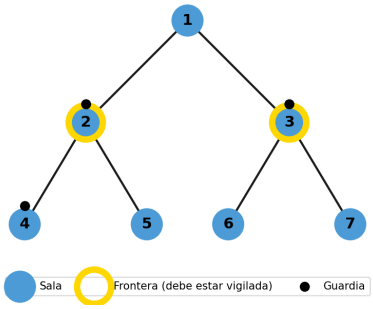


- **Step 3:** ...and simultaneously toward room 3. Now rooms 2 and 3 are the new frontiers that block

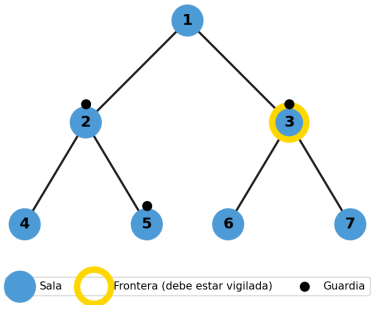
the thieves from escaping. Notice that in this step only 2 active guards are actually required (one at each frontier), with the third one waiting.



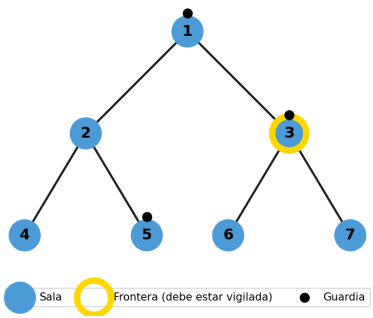
- **Step 4:** Keeping the frontiers secured in rooms 2 and 3, the third guard moves in to clear room 4. This is the critical moment where all 3 guards are needed at the same time!



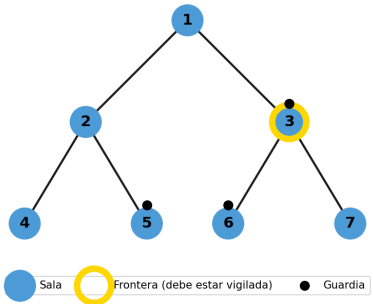
- **Step 5:** The guard finishes checking room 4 and moves to room 5. The left branch is secured.



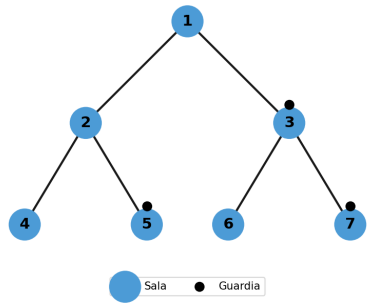
- **Step 6:** Since the left branch is now free of thieves, the guards in that area can leave the frontier at room 2 and move back toward room 1.



- **Step 7:** With the frontier firmly maintained in room 3, the support guard moves forward to clear room 6.



- **Step 8:** Finally, the guard checks room 7. All thieves have been cornered and arrested.



Path to the OIE

AUTOBUSES



Ariadna and Darío travel together to reach the bus station, from where they will take a bus to Valladolid to attend the OIE 2026. To get there, they must go through the metro network.

The metro network consists of n stops, numbered from 1 to n, and m lines. The i-th line has one-way trains going from stop a_i to stop b_i (but not the other way), and the journey takes d_i minutes. We assume transfers are instantaneous. There may be more than one line between two stops, and there may be stops with no incoming or outgoing lines.

Each stop has a color: red or green. When Ariadna and Darío are at a red stop, Ariadna will decide which line (among those leaving that stop) they take. When they are at a green stop, Darío will decide which line they take. If they arrive at a stop from which no lines depart, then they will have to stay there forever and will never reach the station.

Ariadna and Darío must go from stop 1, next to their home, to stop n, the bus station. Once they reach the bus station, they will not take any more metro and their journey will end, regardless of the color of this final stop.

Ariadna likes to arrive early to places to have some margin, so she will try to minimize the total travel time. Darío, who knows he has work waiting for him, prefers to arrive as late as possible, trying to maximize the total time.

Both will make decisions optimally knowing the other's strategy. Note that for Darío, any situation in which they never reach the station (infinite time) is always preferable to any finite time, while Ariadna, on the other hand, prefers any finite time over an infinite one.

How long will it take them to reach the bus station?

Entrada y salida

The first line contains an integer T, the number of test cases.

Then follow T test cases, each with $m + 2$ lines:

- The first line contains n and m, the number of stops and metro lines, respectively.
- The second line contains s, a string of n characters, where the i-th is '`R`' or '`V`' depending on whether the i-th stop is red or green.
- The next m lines each contain 3 integers, a_i b_i d_i, describing the i-th line.

For each test case, print a line with either an integer, the time it takes them to arrive, or the word "`INFINITE`" if it is not possible to reach from stop 1 to the station under any circumstances, or if Darío manages to force them to loop infinitely without reaching the station or get stuck at a stop with no outgoing lines.

Restricciones

- $1 \leq T \leq 10^4$
- $2 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq d_i \leq 1000$

- a_i, b_i are different integers between 1 and n.
- The sum of n over all test cases is at most $2 \cdot 10^5$.
- The sum of m over all test cases is at most $2 \cdot 10^5$.

Subtareas

1. (13 points) All stops are red.
2. (23 points) All stops are green.
3. (15 points) $a_i < b_i$ for all $1 \leq i \leq m$.
4. (24 points) The total sum of n and m over all test cases is at most 2000.
5. (25 points) No additional constraints.

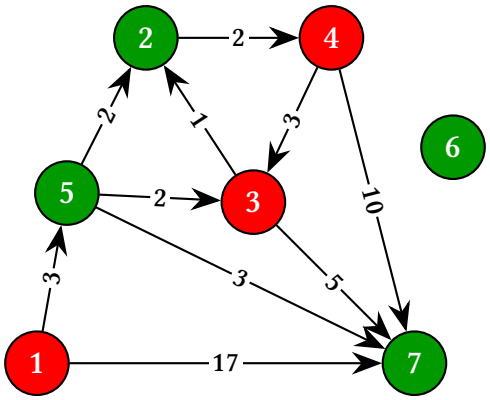
Ejemplo

Entrada

```
3
7 10
RVRRVVV
1 5 3
1 7 17
5 7 3
5 2 2
5 3 2
2 4 2
3 2 1
3 7 5
4 7 10
4 3 3
3 3
VRR
1 2 7
1 2 10
1 3 10
4 6
VRVR
1 2 1
1 4 1
3 2 1
3 4 1
2 1 5
2 3 5
```

Salida

```
15
INFINITO
INFINITO
```



In the **first case**, Ariadna starts choosing from stop 1 (since it is red). She must choose between going to stop 5, taking 3 minutes, or directly to stop 7 (the bus station) and finishing, but that would take 17 minutes. She chooses to go to stop 5, since they will arrive sooner that way and she wants to arrive early.

Now Darío chooses from stop 5 (since it is green). He must choose between going to stop 7 in 3 minutes, to stop 3 in 2 minutes, or to stop 2 in 2 minutes. Since he wants to arrive as late as possible, the first option is not appealing. If he chose to go to stop 3, Ariadna would then go directly to stop 7 and the total duration would be $3 + 2 + 5 = 10$ minutes. Since going to stop 2 can lead to a longer journey, he chooses stop 2.

Now Darío must choose again at stop 2 (since it is green), but he only has one option: go to stop 4 in 2 minutes, so he is forced to take that line.

Now Ariadna chooses from stop 4. Her options are to go directly to stop 7, taking 10 minutes, or to first go to stop 3. She sees that from stop 3 she can go to stop 7 in 5 minutes, so they arrive earlier if she goes to stop 3 first.

Finally Ariadna chooses from stop 3. She can either go to stop 2 in 1 minute or to stop 7 (the bus station) in 5 minutes, and clearly prefers the latter.

They finally reach the bus station, after a journey lasting a total of $3 + 2 + 2 + 3 + 5 = 15$ minutes.

The other two cases are the two ways in which Darío can ensure they never reach the station.

In the **second case**, if Darío chooses to go to stop 2 (by either of the two lines), they arrive at a stop with no outgoing lines, and therefore can never leave.

In the **third case**, if Darío always chooses to go to stop 2, he can force them to loop infinitely without ever reaching stop 4.